PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

JOINT INSPIRING EXAMINATIONS FOR PRIVATE SECONDARY SCHOOLS

(JIEPSS)

FORM THREE JOINT TERMINAL EXAMINATION

CODE 031 **PHYSICS**

Time: 3 Hours **May/ June 2025**

INSTRUCTIONS;

1. This paper consists of three sections A, B and C with total of eleven questions.
2. Answer all questions in sections A, B and only two questions from section C.
3. Non-programmable calculators may be used.
4. Write your examination name on every page of your answer sheet.
5. Whenever necessary the following constants may be used;

- Acceleration due to gravity = $10m/s^2$.
- Density of water = $1g/cm^3$
- Density of air = $1.29kg/m^3$
- Specific heat capacities of water, ice and copper are; 4200J/kgk, 2100J/kgk and 390J/kgk respectively.
- Linear expansivities of iron and brass are; 1.2×10^{-5} /k and 1.9×10^{-5} /k respectively.
- The latent heats of fusion and vapourizations are 3.34×10^5J/kg and 2.2×10^6J/kg

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SECTION A (16 MARKS)

1. Choose the answer from the given alternatives and write its letter beside the item number in the answer sheet provided.

i. A photographer sets up two mirrors at a 60^0 angle to capture unique images. If a subject is placed between the mirrors, how many images of the object will be visible in the reflection?
A. Infinite B. 3 C. 4 D. 6 E. 5

ii. What can be your comment on, in which direction does the light bend as it travel from denser medium to a rare dense medium?
A. Towards the normal
B. Towards the incidence ray
C. Away from the normal
D. Parallel to the normal
E. It maintains its path

iii. Camara kicked the ball during the football match. What can you comment on the scenario basing in Newton's laws of motion?
A. The player's foot get hurt
B. The ball moves because the player's foot exerts force
C. The foot moves behind because of the effect of force
D. The ball stays still
E. The ground exerts a force on the player's foot.

iv. Which physics concept among the followings is used in testing the purity of milk?
A. Osmosis B. Capillarity C. relative density D. permeability E.Upthrust

v. What role does iris play in human eye?
A. To hold the lens in position
B. To prevent internal reflection
C. To control the size of the pupil
D. To control the thickness of the lens
E. To protect eye from light

vi. Which metal can be strongly magnetized when subjected to magnetic fields?
A. Nickel and copper
B. Zinc and aluminium
C. Cobalt and iron
D. Aluminium and lead
E. Iron and zinc

vii. A body is moved upward a distance of 20m. How long will it take to reach the maximum height?
A.2s B. 3s C. 10s D. 5s E. 12s

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viii. Why do people prefer white shawl to wrap around a baby to keep the baby warm?
   A. It is poor conductor
   B. It indicates cleanliness
   C. It is a poor radiator
   D. It has pockets of air trapped in it
   E. It is good reflector of heat

ix. How much heat energy is given out by a copper block of mass 40g as it cools from 1840^0c to 40^0c?
   A. 28008J B. 2800J C. 2808J D. 28000J E. 28080J.

x. If you increase the magnification of the objective lens in a compound microscope, what happens to the fields of view?
   A. IT becomes smaller
   B. It becomes larger
   C. It remains the same
   D. It becomes blurry
   E. It becomes colourless

2. Match the items in list A with the items in list B by writing its letter beside the item number in the answer sheet provided;

LIST A	LIST B
i. Magnetic materials ii. Law of polarity iii. Magnetic shielding iv. Neutral point v. Alumium vi. Angle of inclination	A. Angle between the resultant earth fields and magnetic equator B. Like poles attracts , unlike repels C. Redirects magnetic lines of force D. Strong magnet E. Magnetic fields is zero F. Paramagnetic G. Diamagnetic H. Direct neutral point I. Iron nail J. Angle between geographical poles and earth's magnetic poles

SECTION B (54 MARKS).

Answer all questions in this section

3. (a). What can be your comment as a colour specialist on the appearances of;
   i. A blue flag when viewed during a day through a sheet of red glass?
   ii. A man wearing blue shirt and red trouser, holding a green coloured hanger chief when viewed in pure yellow light?

   (b). An optical arrangement consists of two convex lenses of focal lengths of 12cm and 6cm respectively for the objective and eyepiece lenses. The instrument is set such that the final image is at infinity. If the object is placed at 24cm in front of the objective lens, calculate the distance of separation between the lenses.

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4. (a) i. The atmospheric pressure is a result of total weight of the air layers on The earth's surface. Briefly explain why it does not crush us as we know its value is 1.03×10^5Pa?

ii. A mercury barometer reads 75cmHg at the base of the mountain that is 5000m high. If the density of mercury is 13600kg/m^3, calculate the barometer reading at the top of the mountain.

(b). A uniform beam of 2m long is supported horizontally by two vertical strings X and Y placed at 20cm and 40cm from ends respectively. If the beam weighs 40kg and it carry as mass of 20kg that hanging at 30cm from the second end of the beam, calculate the tensions on the strings.

5. (a). Two machines, which are combination pulleys and wheel and axle were compared and it was found that when used in lifting a load of 5000N they have the same efficiency. The radius of the wheel is 50cm and that of theaxle is 5cm. If the combination pulley has 5 pulleys in the system, comment on the effort applied at the wheel and axle with respect to that applied to the combination pulleys.

(b). i. State the principle of conservation of linear momentum.

ii. If a body loses half of its velocity on penetrating a 4cm wooden block, how much more will it penetrate before coming to rest?

6. (a). By using kinetic theory of matter, briefly describe the reason for the particles in solid to expand when placed in boiling water.

(b). A piece of copper material of mass 400g at 100^0c was quickly transferred into a 40g of water at 40^0c. Find the final temperature of the mixture.

7. (a). A magnetic material can be strongly added with the attractive property sothat it can be acting as a magnet that is permanently made. Describe three ways in which that can be effectively made possible.

(b). A 1000kg car travelling at 15m/s has its speed increased to 45m/s. How many times greater is the kinetic energy at final velocity than that at initial velocity.

8. (a). The geothermal energy can be extracted and brought in use in terms of electricity. Explain how its reservoirs originated.

(b).i. Form two students started arguing about easily found liquid in their environment which is water not being used as a thermometric liquid. As a form three student how can you solve their arguments by comparing the easily available liquid and the reasonable thermometric liquid?

ii. The length of the mercury column in a non-calibrated mercury thermometer is 2cm when its bulb is immersed in melting ice and 20cmwhen the bulb is in steam above boiling water. What would the temperature be if the length of mercury column becomes 11cm in a given liquid?

SECTION C (30 MARKS)

Answer only two (2) questions in this section.

9. (a). Scalar and vector quantities can be different in considering their operations. Scalar can be operated using normal arithmetic operations, but in vector operations rules are to be applied. State those rules.

(b). An airplane flying due to east at a velocity of 400km/h, its velocity relative to the air, while it encounters a wind moving at 200km/h blowing 45^0 east. What is the velocity of airplane according to an observer on the ground?

(c). The force that oppose motion is guided by several laws. Using examples state three of them.

(d). A box of mass 40kg was pulled along a flat plane inclined at 60^0 to vertical. If the plane is 4m long, calculate the work done in moving the box up the top of the plane.

10. (a). Submarines are sinking and floating ships? Using diagrams explain the working mechanism of the submarine as used in war. Hence state the law helped you to give your description.

(b). An object with a volume of $150cm^3$is found floating in water with 60% of its volume submerged, what is the density of the object?

(c). $100cm^3$of water are mixed with $800cm^3$of brine of density $1027kg/m^3$. Calculate the density of the mixture.

(d). Calculate the heat energy absorbed by 30g of ice at -10^0c when heated to change it to water at 90^0c.

11. (a). An old woman was asked by her grandson to help him ironing his trouser since he was finishing holiday package as he was to be taken back to school next day. She get amazed on how electric iron get heated when connected to the power supply. As an expected electrical engineer describe to her factors affecting the heating effect of an electric iron as it is connected to the supply.

(b). A battery is connected to a 10Ω resistor and a switch in series. A voltmeter is connected across the battery. When the switch is open the voltmeter reads 1.45V and 1.26V when the switch is closed. Comment on the internal resistance of the battery?

(c). A house of four bed rooms each installed with 5 bulbs rated 60W, 240V. They are left on from 7:00pm to 10:30pm every day. If the electrical company charges 357 tshs per unit, comment on the cost of running the house for a month of 28days.

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